

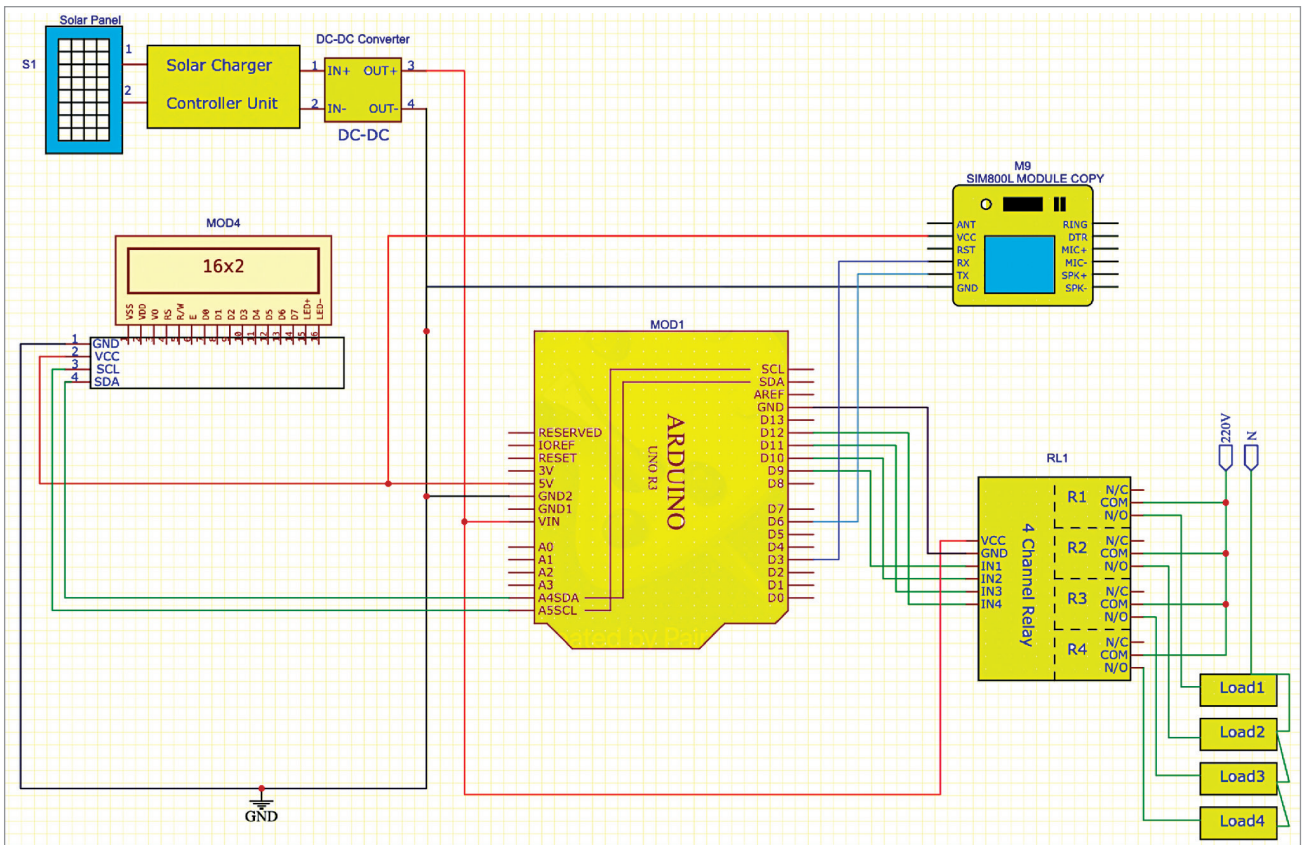


Fig. 2: Circuit diagram

```

1 #include <Wire.h>
2 #include <LiquidCrystal_I2C.h>
3 #include <SoftwareSerial.h>
4 LiquidCrystal_I2C Lcd(0x27,16,2); //set the lcd address to 0x27 for a 16 chars and 2 line
5 SoftwareSerial mySerial(3, 6);
6 String task;
7 void setup()
8 {
9   Lcd.init(); //initialize the lcd
10  Lcd.init();
11  Lcd.backlight(); // Turn on the backlight and print message.
12  mySerial.begin(9600); // Setting the baud rate of GSM Module.
13  Serial.begin(9600); // Setting the baud rate of Serial Monitor (Arduino). pinMode(9,OUT
14  pinMode(10,OUTPUT);
15  pinMode(11,OUTPUT);
16  pinMode(12,OUTPUT);
17  digitalWrite(9,HIGH);
18  digitalWrite(10,HIGH);
19  digitalWrite(11,HIGH);
20  digitalWrite(12,HIGH);
21  delay(1000);
22  SendMessage();
23  Lcd.clear();
24  delay(500);
25  Lcd.print(" by Yogesh jatav");
26  delay(1000);
27 }
28 void loop()
29 {
30   if (Serial.available()>0)
31   switch(Serial.read())
32   {
33     case 's':
34       SendMessage();
35       break;
36     case 'r':
37       RecieveMessage();
38       break;

```

Fig. 3: Code snippet setting of the baud rate, display address

in the code; here, the project uses 16x2, so that setting is applied. Software serial pin and baud rate need to be configured in the code. The default baud rate of the SIM800L GSM module is 9600, which is utilised here.

Construction and testing

Assemble the circuit on a general-purpose PCB or a breadboard. After assembly, enclose it in a suitable casing.

Before assembly, upload the source code into the Arduino Uno board. Insert the SIM card with a message pack into the GSM module. Power the device and wait until the GSM module connects to the network.

Now, one can send messages like 'Switch2 off' or 'all on' (commands as set in the code) to the device and control the AC motor, pump, or other devices connected to the relay module. **EFY**

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